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Momentum-growing neutrino–quark interactions and the KM3NeT event: an analytic and public-data audit (v0.9)

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Abstract: We revisit a dimension-seven momentum-growing neutrino–quark interaction after the observation of KM3-230213A and the 12.6-year IceCube extremely-high-energy (EHE) search. A leading-order parton calculation with the conservative partonic-energy requirement $\sqrt{\hat{s}} < \Lambda$ retains order-one cross-section changes at hundreds of PeV for a collider-bound coefficient. We derive an exact common-flux normalization identity and audit the detector topology: the Case-6 operator has a neutral final state, whereas the published KM3NeT response selects a bright muon track. We validate the released KM3NeT Ehlert-spectrum benchmark to 0.006% and use IceCube's released differential-limit curve in its stated sliding one-decade E^{-1} windows. In an explicitly labelled all-flavor response proxy, the limiting common-flux scales are 0.155, 0.128, 0.102 and 0.0747 for $\Lambda = 1.5, 2.0, 2.44, 3.0$ TeV. The corresponding unmodified direct bright-track expectations are 0.0376, 0.0311, 0.0249 and 0.0181 events. These are screening diagnostics, not frequentist 90% limits: the public release lacks the reconstructed-bin information and beyond-Standard-Model response needed to reproduce the published likelihood. We identify the required topology and transport calculations for a definitive test and state precise conditions under which alternative interactions could evade the common-flux suppression.

Keywords: Neutrino interactions, effective field theory, cosmic neutrinos, IceCube, KM3NeT

Contents

1 Introduction and scope **1**

2 Literature and novelty **2**

2.1 Earth transport and neutrino deep-inelastic scattering 2

2.2 The operator, collider information and cosmic flux 2

2.3 Recent IceCube and KM3NeT comparisons 2

3 Analytic structure of Case 6 **3**

4 Public data and controlled screen **4**

5 Results and interpretation **4**

6 Which alternatives remain viable? **5**

7 Reproducibility and limitations **6**

Contents

1 Introduction and scope

The KM3NeT collaboration reported a very bright track, KM3-230213A, made by a muon traversing the partially built ARCA array. For an E_ν^{-2} incident spectrum its median parent-neutrino energy is about 220 PeV and its stated central 90% range is approximately 72 PeV–2.6 EeV [1]. The event invites a quantitative comparison with IceCube's search for neutrinos above a few PeV, which found three PeV-scale events in 12.6 years but no event well above 10 PeV [2]. The question here is narrower than explaining the origin of one event: can the momentum growth of a laboratory-constrained neutrino–quark operator increase an ARCA bright-track expectation after the same flux passes the IceCube test?

We use an Ehlert-type cosmogenic spectrum as a fixed shape, with a single free normalization κ [3]. The leading-order cross-section table and the published detector effective areas are kept separate. This makes the structural result independent of the numerical integral, while exposing the point where topology and response information are missing. Version 0.9 is a reproducible public-data audit. It does not claim to implement the collaboration's event-level EHE likelihood.

2 Literature and novelty

2.1 Earth transport and neutrino deep-inelastic scattering

Tau-neutrino propagation differs from pure absorption because charged-current production of a tau followed by decay repopulates the neutrino flux at lower energies. Dutta, Reno and Sarcevic developed the coupled propagation and decay treatment and quantified the dependence on the injected spectrum [4]; their later study showed that secondary muon neutrinos from tau decay can contribute to upward-muon rates, with a particularly strong spectral-shape dependence [5]. Rakshit and Reya studied iterative solutions of coupled ν_τ–τ transport with secondary ν_μ production and convergence checks [6]. These works set the appropriate standard for a future topology-aware Case-6 propagation calculation. The inherited v0.3 transport implementation was checked against these historical benchmarks, but the present EHE fold does not rerun that transport with the new differential kernel.

Modern Standard-Model neutrino DIS predictions include higher-order QCD, small-x PDF extrapolation and nuclear uncertainties. Xie *et al.* provide CT18-based predictions over the UHE range and quantify those effects [7]. Our CT10nlo-based leading-order parton integral is a controlled *ratio* benchmark for the new operator; it is not a modern precision Standard-Model cross-section calculation. PDF, perturbative and nuclear uncertainties must be revisited before a precision exclusion claim.

2.2 The operator, collider information and cosmic flux

Pandey, Karmakar and Rakshit studied monojet and neutrino constraints on a range of non-standard interactions, including their Case 6, a derivative dimension-seven neutrino–quark contact interaction. Their quoted monojet bound for its coefficient is $|C_4| \lesssim 8.6 \times 10^{-10}\,\mathrm{GeV}^{-3}$ [8]. The earlier work compared cross sections and collider limits, while the present audit combines its momentum-growing benchmark with the subsequently released 12.6-year EHE response, the observed energy-dependent limit and the specific ARCA bright-track topology. We use the quoted coefficient as a benchmark, not as an EFT-valid recast of LHC events after partonic truncation. A UV completion and a collider analysis with the same validity cut would be required to make that correspondence fully consistent.

Ehlert *et al.* constrained high-energy proton fractions using multimessenger cosmic-ray information and provided cosmogenic-neutrino spectra that depend on source evolution and acceleration assumptions [3]. Choosing one of these shapes fixes the comparison, but does not remove the astrophysical uncertainty in its normalization or composition. The KM3NeT open analysis notebook supplies a sky-averaged, all-flavor bright-track effective area and per-flavor $\nu + \bar{\nu}$ flux convention [9]. The full ARCA performance study treats track and shower selections separately [10], which is essential when the interaction has a neutral final state.

2.3 Recent IceCube and KM3NeT comparisons

IceCube's EHE analysis fits reconstructed energy and direction with a binned Poisson likelihood, profiles an astrophysical component, and calibrates tests with pseudoexperiments [2]. Its release gives true-energy effective areas and the published quasi-differential limit,

but not the full reconstructed-bin response [11]. KM3NeT's broader comparison of its event with the global neutrino landscape uses joint flux fits [12]. A later analysis used the single KM3NeT event to constrain the neutrino–nucleon cross section, subject to its flux and detector hypotheses [13]; IceCube also searched for point-source emission in the event direction [14]. Those studies address complementary questions. Our specific novelty is the joint analytic flux-ratio and topology audit of the Case-6 operator with both published public responses. The result is conditional on a stationary isotropic common flux and cannot exclude a transient or anisotropic source using this all-sky calculation.

3 Analytic structure of Case 6

The schematic low-energy interaction is

$$\mathcal{L}_{\rm C6} = C_4\,(\bar{\nu}_i^c \overleftrightarrow{\partial}_\mu \nu_j)(\bar{q}\gamma^\mu q) + \text{h.c.}, \qquad [C_4] = -3. \tag{3.1}$$

The flavor and charge-conjugation assignments in a concrete theory determine which incident species and regenerated species are affected. The key topology selection rule is robust: the hard scattering in eq. (3.1) produces a neutrino-sector final state and hadrons, not a charged muon. It adds no direct $\nu_\mu N \to \mu X$ track channel. For an orthogonal neutral final state the new amplitude contributes a non-negative additive term rather than interference with the Standard Model charged-current track amplitude.

For a nucleon at rest and Bjorken variables x, y,

$$\hat{s} \simeq 2m_N E_\nu x, \qquad Q^2 = \hat{s}y, \qquad \overline{|\mathcal{M}_{\rm C6}|^2} = 64 C_4^2 \hat{s}^3 y(1-y). \tag{3.2}$$

The polynomial vanishes at $y = 0, 1$ and peaks at $y = 1/2$; the $\hat{s}^3$ factor explains the strong energy growth before PDF convolution. The primary EFT restriction is $\sqrt{\hat{s}} < \Lambda$, imposed point by point. The weaker $Q < \Lambda$ restriction admits configurations with $Q^2 = \hat{s}y < \Lambda^2$ while $\hat{s} > \Lambda^2$ and is reported only as an aggressive envelope. The dimensionless coefficient $c_4 = C_4\Lambda^3$ obeys $c_4/(4\pi) = 0.231, 0.547, 0.994, 1.848$ at the four cutoffs used below. Thus $\Lambda = 3$ TeV crosses the simple 4π coefficient diagnostic, and 2.44 TeV sits near it. These diagnostics are not a complete partial-wave unitarity analysis. The expansion parameter is $\hat{s}/\Lambda^2$; omitted higher-dimensional terms are uncontrolled near the imposed boundary, so even the primary cut should be interpreted as an EFT-domain envelope rather than a precision UV prediction.

Let I and K denote the predicted IceCube and KM3NeT counts per unit common flux normalization, with the full energy, angle and topology response understood. Then the exact identity

$$N_{\rm IC}(\kappa) = \kappa I, \quad N_{\rm KM3}(\kappa) = \kappa K, \quad \kappa \le \frac{U}{I} \quad \Longrightarrow \quad N_{\rm KM3} \le U\frac{K}{I} \tag{3.3}$$

holds for any constraint of the indicated form. If a proposed interaction enhances I and K by similar factors, K/I barely changes. This explains the near-cancellation seen in the initial v0.8 screen. For an actual track sample, Case 6 changes K only indirectly through propagation, flavor conversion or secondary-muon production; none is quantified by simply multiplying its published track area by $1 + r$.

4 Public data and controlled screen

For equal incident per-flavor fluxes $\phi_{\rm pf}$ and the IceCube release's flavor-summed area, the Standard-Model event fold is

$$N_{\rm IC}^{\rm SM} = 4\pi T_{\rm IC} \int dE\, \phi_{\rm pf}(E) A_{\rm tot}^{\rm SM}(E). \tag{4.1}$$

The area is supplied in m^2 and is multiplied by 10^4 for use with a flux in cm^{-2}. Its columns satisfy $A_{\rm tot} \simeq A_e + A_\mu + A_\tau$ to tabulation precision. The analogous KM3NeT fold uses its released all-flavor bright-track area and $T_{\rm KM3} = 335$ days. We use $T_{\rm IC} = 12.6$ Julian years.

To expose the topology uncertainty we compute two IceCube response *proxies*, never treating them as a re-simulation of the detector:

$$A_{\rm all}^{\rm proxy}(E) = A_{\rm tot}^{\rm SM}(E)[1 + r(E)], \tag{4.2}$$

$$A_{\tau\text{-init}}^{\rm proxy}(E) = A_{\rm tot}^{\rm SM}(E) + A_\tau^{\rm SM}(E) r(E), \qquad r = \sigma_{\rm C6}/\sigma_{\rm SM}. \tag{4.3}$$

These bracket choices about the incident flavors being affected, but they do not bound the true detector response: neutral-current visible energy, Earth attenuation and regeneration may change the selection in either direction. The published KM3NeT $A_{\rm eff}$ is for bright tracks. We therefore keep the direct Case-6 contribution to that response exactly zero. Its hypothetical $A_{\rm KM3}^{\rm SM}(1+r)$ fold is retained in the machine-readable table solely to show the size of a *wrong-topology* acceptance proxy.

IceCube's released differential-limit ordinate L_j is an all-flavor $E^2\Phi$ at central energy E_j. The collaboration constructs each point using an E^{-1} test spectrum across a sliding one-decade window [2]. We reconstruct its true-energy reference flux as

$$\phi_{j,\rm pf}^{\rm ref}(E) = \frac{L_j}{3E_jE}\,\Theta(E - E_j/\sqrt{10})\Theta(\sqrt{10}E_j - E), \qquad n_j^{\rm ref} = 4\pi T_{\rm IC} \int dE\, A_{\rm tot}^{\rm SM} \phi_{j,\rm pf}^{\rm ref}. \tag{4.4}$$

For a chosen model response proxy, define $R_j = n_j^{\rm model}/n_j^{\rm ref}$ using the same window, and the diagnostic scale

$$\kappa_{\rm window}^{\rm proxy} = \min_j R_j^{-1}. \tag{4.5}$$

This uses the *observed* limit rather than the no-signal sensitivity. Overlapping windows are correlated; the published limit is calibrated for its E^{-1} reference and Standard-Model response. Equations (4.4)–(4.5) do not inherit 90% coverage for a broad Ehlert spectrum or modified cross section. The published full likelihood would require event counts and signal/background templates in reconstructed energy and direction, a migration response and the nuisance treatment. These are absent from the four-file public release. Thus a genuine likelihood replacement cannot be produced from the released tables alone.

5 Results and interpretation

The frozen v0.8 cross-section grid spans 0.1–20,000 PeV and uses CT10nlo with the pointwise $\sqrt{\hat{s}} < \Lambda$ cut. Its no-cut ratio $r(0.1\,\mathrm{PeV}) = 5.34 \times 10^{-4}$ and $r(1\,\mathrm{PeV}) = 1.38 \times 10^{-2}$ is

Table 1. Version 0.9 public-data diagnostics. $\kappa_{\rm all}$ and κ_τ are the window-screen scales of eq. (4.5), not confidence limits. The last two columns are the *unchanged direct* bright-track expectation at those scales; propagation effects have not been evaluated.

Λ [TeV]	$c_4/4\pi$	$N_{\rm IC}^{\rm all}$	$N_{\rm IC}^{\tau}$	$\kappa_{\rm all}$	$N_{\rm KM3}^{\rm tr}(\kappa_{\rm all})$	$N_{\rm KM3}^{\rm tr}(\kappa_\tau)$
1.50	0.231	20.734	18.885	0.1549	0.0376	0.0413
2.00	0.547	25.051	20.608	0.1280	0.0311	0.0378
2.44	0.994	31.151	23.049	0.1025	0.0249	0.0338
3.00	1.848	42.537	27.637	0.0747	0.0181	0.0281

preserved as a historical regression. The v0.9 numerical work only folds those saved curves with the public data. The KM3NeT Ehlert expectation is 0.242754 events, versus 0.242767 from its official notebook; in the published central 90% energy range it is 0.211081 versus 0.211117. The corrected IceCube area fold is 17.6748 events at the unscaled Ehlert benchmark. The difference from the v0.8 value 17.6746 is the first numerical row of its headerless area file, which the old parser silently skipped. The original v0.8 unit error gave 0.001767 before the $\mathrm{m}^2 \to \mathrm{cm}^2$ repair.

For all four benchmarks the most constraining released window is centered at $E_j = 3.162 \times 10^8$ GeV, or about 316 PeV. The all-flavor window ratios there are 6.46, 7.81, 9.76 and 13.38, respectively. The published reference yield in that window is approximately 2.42 events when folded with the released true-energy area. These numbers show why the unscaled cosmogenic benchmark is strongly disfavored by the public energy-dependent screen. They do not measure a frequentist model-rejection probability.

The reported $N_{\rm IC}^{\rm all}$ in table 1 uses an all-flavor $1 + r$ acceptance proxy. If only tau incident states are affected, using A_τ reduces the effect as shown by $N_{\rm IC}^{\tau}$; the corresponding κ_τ values are 0.1703, 0.1559, 0.1391 and 0.1157. This flavor dependence is another reason that the v0.8 single-column proxy cannot be interpreted as a physical event prediction. In particular, the v0.8 $N_{\rm KM3}^{\rm NC\ equivalent}$ enhancement is a diagnostic of what happens if an NC cross section is assigned a track acceptance, not a mechanism for the observed muon. The direct track count at fixed incident flux is the Standard-Model count in this calculation.

6 Which alternatives remain viable?

Equation (3.3) gives a necessary quantitative target for variants. At $\Lambda = 2$ TeV, the all-flavor screen leaves about 0.031 direct track events, so a model seeking one expected bright track at the same screened common flux must raise the *track-to-IceCube yield ratio* by roughly $1/0.031 \simeq 32$. This is a necessary-condition diagnostic, not a model-independent exclusion: the IceCube constraint and propagation must be recomputed for each model. The following distinctions matter.

An interaction with a charged-current muon final state can create tracks directly, but the same interaction changes Earth opacity and IceCube's tracks. Its laboratory bounds, gauge embedding and EFT validity must be derived before any detector comparison. A neutral-current operator coupled only to tau neutrinos can change tau regeneration or

generate secondary muon neutrinos through subsequent tau decays; this requires the coupled transport kernels of [5, 6] with the actual Case-6 differential inelasticity distribution and flavor assignment. Directional or time-dependent sources can break the common isotropic exposure ratio; the present sky-averaged areas are inadequate for such models. Energy-localized or resonant new physics can also change the IceCube/ARCA response ratio, but a contact EFT cannot be extrapolated through its resonance, and the UV completion must obey laboratory and unitarity constraints. These variants are research paths, not positive solutions demonstrated here.

7 Reproducibility and limitations

The project records v0.7 and v0.8 failed and corrected stages in `PROGRESS.log`; v0.9 inputs and outputs are under versioned `data/uhe_official` and `results/v0.9`. The one-dimensional quadrature uses 6001 logarithmic energy points between 10^5 and 10^{11} GeV. The released effective areas are linearly interpolated as in the collaboration's benchmark notebook, and the tabulated Ehlert $E^2\phi$ is interpolated in its provided logarithm. The quadrature is cheap; parallelizing it over 14 cores would add overhead and risk oversubscription. The expensive LHAPDF grid is reused rather than recomputed. Automated checks enforce area units, flavor-column sums, and the two KM3NeT benchmark integrals.

The largest limitation is detector response under new physics. The published effective areas include selection and Standard-Model Earth effects. Multiplication by a cross-section ratio does not account for neutral-current visible energy, energy migration, track/cascade classification, flavor-changing propagation or angle dependence. The second limitation is statistical: the released differential limits are not the model-specific binned likelihood. Third, CT10nlo and leading-order DIS are not precision inputs at all energies, and the 4π check is only a perturbative diagnostic. A definitive Case-6 test needs collaboration-level response matrices or event simulation, a validated modified Earth-transport calculation, and a consistently truncated collider/UV interpretation. Until then the v0.9 conclusion is that momentum growth survives in the cross-section calculation, while its naive bright-track explanation is unsupported and the common-flux event-yield proxies are strongly constrained by IceCube.

AI-assisted drafting disclosure

This working manuscript and its reproducibility scripts were prepared with AI assistance under the author's direction. Bibliographic records and quoted public-data conventions were checked against the primary papers and collaboration releases listed above. The author should independently verify every numerical and scientific claim before journal submission.

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